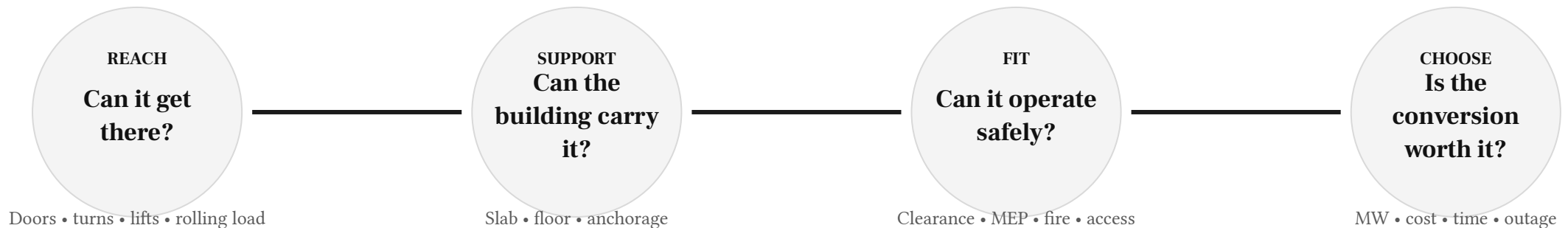


Check whether an existing hall can really become AI-ready—before committing capital

Before the owner promises an AI-ready hall, we check whether the equipment can get into the building, whether the floor can safely carry it, whether everything fits and what must be changed.



Can it get there?

Can it fit?

Can the building carry it?

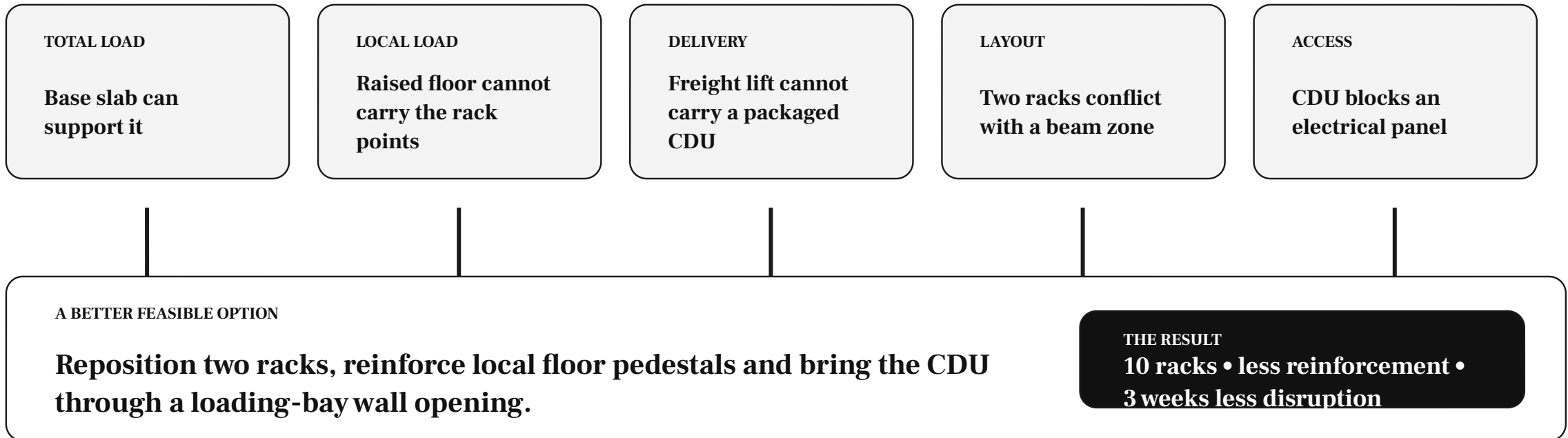
Is the conversion worth it?

For engineering, design, development and portfolio-capital teams deciding which hall, layout and enabling work should proceed.

The total weight looks acceptable—but the proposed conversion still does not work

PROPOSED AI HALL

Twelve 130 kW racks and two CDUs in an existing raised-floor hall.



The governing constraint is the minimum of structure, space, delivery, power, cooling, life safety and operations—not the average capacity.

An engineer-reviewed conversion decision—not a binary structural answer

FEASIBLE WITH ENABLING WORK

Hall B can support 1.30 MW with a revised layout.

Governing constraint	Local raised-floor capacity at rack positions
Layout change	Relocate two racks away from the beam zone
Delivery solution	Use loading-bay wall opening with temporary works
Indicative enabling work	£190,000 and a one-week outage
Confidence	High—subject to licensed engineer approval

DECISION VIEW

●	Capacity	Feasible MW and rack density
●	Blocker	Located governing constraint
●	Intervention	Layout, reinforcement or reroute
●	Delivery	Route, rigging and temporary works
●	Disruption	Outage and live-site impact
●	Investment	Cost, time, value and confidence

The product screens and compares options. Licensed engineers retain structural adequacy, reinforcement, anchorage and construction approval.

Stop impossible conversions early—and fund the hall that unlocks capacity sooner

TODAY

Retrofit decisions begin with power and cooling; physical constraints arrive later.

- Drawings are treated as current reality
- Total floor load hides local failure
- The equipment layout ignores logistics
- A feasible design can disrupt live service
- Each hall is studied as a separate project

WITH CONVERSION PLANNING

Every hall is tested against one integrated feasibility and investment view.

- ✓ Expose what is verified, uncertain or survey-required
- ✓ Check point, rolling, beam-zone and raised-floor constraints
- ✓ Test packaging, doors, turns, lifts, staging and temporary works
- ✓ Compare outage, constructability, tenant and work-window effects
- ✓ Rank MW, capex, time, disruption and confidence across the portfolio

Practical value: faster feasibility, less wasted design, earlier blocker discovery, lower disruption and better capital allocation.

Join building reality, engineering constraints and commercial choices in one model

THE DATA WE NEED

1	Building structure	Drawings • calculations • known changes
2	Current reality	LiDAR/scan • survey • existing equipment
3	Proposed deployment	Rack/CDU weights • footprint • clearance
4	Delivery + operation	Routes • lifts • rigging • maintenance access
5	Integrated services	Power • cooling • MEP • fire • egress
6	Investment choices	Cost • schedule • outage • capacity value

HOW THE ANSWER IS BUILT

- 1 Establish what is real**
Reconcile drawings, models, scans and visible conditions
- 2 Screen fit, route and support**
Check loads, openings, turns, clearance and anchorage
- 3 Propagate every constraint**
Test MEP, power, cooling, life safety and live operations
- 4 Compare feasible interventions**
Rank layout, reinforcement and reroute by MW, cost and outage

A scan proves geometry—not structural capacity. Every conclusion carries its evidence type, accuracy, uncertainty and required professional review.

What must be true before we test the use case

Start with three to five halls and one real rack/CDU configuration—not a generic density target or portfolio-wide build.

✓	A real deployment brief	Operating and shipping weights, footprint and clearances
✓	Credible building evidence	Structural/architectural records or permission to survey
✓	Current loads ♦ routes	Existing equipment, raised floor, openings, lifts and access
✓	Integrated constraints	Preliminary power, cooling, MEP, life safety and outage limits
✓	Professional review	A licensed structural engineer validates material conclusions

If structural evidence cannot be established and survey is refused, deliver a bounded engineering study—not an automated product.

Plain language on the left; technical capability on the right

WHAT THE CLIENT HEARS	WHAT WE BUILD	WHAT IT DOES
● Read drawings and building records	Drawing intelligence	Extract grids, members, dimensions, levels, notes and ratings
● Compare records with the real hall	Model reconciliation	Find differences between drawings, BIM, scans and current geometry
● Check whether equipment can reach and fit	Geometry reasoning	Test envelopes, turns, openings, clearance and delivery routes
● Screen whether it can be supported	Deterministic calculations	Apply engineer-approved point, rolling, distributed and anchor checks
● Find what truly governs the layout	Constraint graph	Propagate structural, spatial, MEP and operational constraints
● Choose the best hall and intervention	Portfolio optimisation	Rank capacity, cost, time, outage, value and confidence

Control boundary: no autonomous structural certification, load rating, reinforcement design, anchorage approval or construction release.